Manaswini Ganjam

mganjam@ncsu.edu

PROFESSIONAL SUMMARY

Motivated to apply geospatial expertise and modeling tools to assess climate change impacts on forest carbon dynamics and ecosystem resilience.

EDUCATION

North Carolina State University — Raleigh, NC
Ph.D. in Forestry and Environmental Resources — Expected May 2025

Jawaharlal Nehru Technological University — Hyderabad, India
M. Tech in Environmental Geomatics — August 2015
B. Tech in Electrical and Electronics Engineering — May 2013

TECHNICAL SKILLS

- R
- Python
- PostgreSQL
- GEE / AWS (Geospatial cloud data)
- Arc GIS/ QGIS
- C, C++

EXPERIENCE

North Carolina State University — Raleigh, NC
Research Assistant — May 2021 – Expected May 2025

- Modeled forest carbon dynamics and resilience using Biome-BGC MuSo, assessing the influences of climate change (RCP 4.5, 6.0, 8.5), stand age, and management scenarios across North American biomes.
- Calibrated and validated ecosystem process-based model simulations with flux tower data, projecting climate-driven shifts in Net Ecosystem Productivity.
- Built a geospatial database synthesizing FIA, NLCD, gSSURGO, and NASA-NEX-GDDP climate projections for the southeastern U.S., applying statistical methods to ensure data integrity.
- Created visualizations (Python, R) and briefed interdisciplinary teams on carbon flux trends, informing sustainable forest management strategies.
- Contributed to multiple research publications and presentations, showcasing ability to share research outcomes with the scientific community.
- Advanced Python, R, and PostgreSQL skills for large-scale geospatial analysis and model parameterization, adaptable to landscape-scale tools.

Central Coffee Research Institute — Bengaluru, India
Project Scientist — June 2017 – March 2018

- Performed pre-processing of high spatial resolution remote sensing data (1m).
- Conducted multiple fieldwork trips for GPS-based ground truth collection.
- Utilized object-based classification for the classification and interpretation of remote sensing data.
- Contributed to the generation of a coffee area map for India.

CSIR-National Geophysical Research Institute — Hyderabad, India
Project Assistant III — November 2015 – March 2017

- Requisitioned satellite datasets like land use, elevation, and vegetation density.
- Digitized high resolution bathymetry data.
- Conducted preprocessing of data, database creation, ancillary data collection.
- Inundation results were obtained using a numerical tsunami model.
- Conducted literature review to assign weights to the parameters using analytical hierarchy process.
- Generated vulnerability maps for Car Nicobar Islands in Indian ocean.

National Remote Sensing Centre Hyderabad, India
Project intern September 2014 – October 2015

- Acquired and managed multi-temporal remote sensing datasets, including Landsat and AWIFS imagery.
- Performed comprehensive data preprocessing, database creation, and ancillary data collection for geospatial analysis
- Implemented unsupervised classification techniques to assess geographical areas affected by forest fires, deforestation, and degradation.
- Conducted in-depth geospatial and statistical analyses to derive meaningful insights from remote sensing data.
- Contributed to the writing and publication of research articles, while actively collaborating on related research projects resulting in multiple peer-reviewed publications.

PUBLICATIONS

- **Ganjam, M.,** Baker, J.S., Aguilos, M., Merganiˇcová, K. (2024). "Maximizing Net Ecosystem Productivity: Integrating Chronosequence Analysis and Ecosystem Modeling Across Contrasting Biomes in North America". Forest Ecosystems (In Revision).
- Fuller, M. R., **Ganjam, M.,** Baker, J. S., & Abt, R. C. (2024). "Advancing forest carbon projections requires improved convergence between ecological and economic models". Carbon Balance and Management.
- Reddy, C.S., Jha, C.S., **Manaswini, G.,** Alekhya, V.V.L.P., Vazeed Pasha, S., Satish, K.V., Diwakar, P. G., & Dadhwal V,K. "Nationwide assessment of forest burnt area in India using Resourcesat-2 AWiFS data". Curr. Sci (2017).
- Reddy, C.S., **Manaswini, G.,** Jha, C.S., Diwakar, P.G., & Dadhwal, V.K. "Development of national database on long-term deforestation in Sri Lanka". Journal of the Indian Society of Remote Sensing, 1-12 (2016).
- Reddy, C.S., **Manaswini, G.,** Satish, K.V., Sonali Singh., Jha, C.S., Dadhwal V,K. "Conservation priorities of forest ecosystems: Evaluation of deforestation and degradation hotspots using geospatial techniques". Ecological Engineering volume 91 (2016):333–342.
- **Manaswini, G.,** and Reddy, C,S., "Geospatial monitoring and prioritization of forest fire incidences in Andhra Pradesh, India". Environmental monitoring and assessment 187.10 (2015):1-12.
- Jakkula, M., Guruhappa, H., **Manaswini, G.,** & Srivastava, K. "Tsunami forces acting on ocean structures: A synthetic study". Journal of Indian Geophysical Union, 21(6), 482-489 (2017).